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Stabilizing high-efficiency perovskite solar cells by homogenizing inter-grain recombination

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Abstract

The long-term stability of polycrystalline perovskite solar cells (PSCs) remains a major hurdle despite their high power conversion efficiencies (PCEs), underscoring the importance of eliminating all potential photodegradation pathways. While efforts have been devoted to address bulk and interfacial defect-induced photodegradation, those correlate to the microscale inter-grain carrier reactions intrinsic to the polycrystalline nature of PSCs remain unsolved. Here we uncover a defect-dependent asymmetric inter-grain recombination pathway that governs photodegradation in mixed-facet PSCs. By resolving the nature and evolution of facet-specific defects for the first time, we reveal that hetero-facet grain boundaries induce non-equilibrium hole accumulation at (111)-facet grain boundaries that unexpectedly accelerates their photodegradation, despite their superior intrinsic photostability. By introducing 3-fluorophenylethylammonium bromide that targets identified formamidinium vacancy defects at the (111)-facet grain boundaries, we homogenize the inter-grain recombination, yielding PSCs with PCEs approaching 26.55% and T90 lifetimes over 2200 h under continuous 1-sun illumination.

Keywords Asymmetric inter-grain recombination, Facet-dependent defects, Grain boundary passivation, Operational stability

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1 Introduction

The operational instability of metal halide polycrystalline perovskite photovoltaics remains a major barrier to their commercialization [1–5], despite high certified power conversion efficiencies (PCEs) for single-junction devices exceeding 27% or tandem devices approaching 35% [6]. While extensive research has focused on defect identification and passivation at bulk and interfacial regions, these macroscopic strategies provide only partial improvements in stability [7–13]. Current strategies often overlook the microscale heterogeneity inherent to polycrystalline perovskite films [14–17], where facet-specific defect dynamics and hidden grain boundary interactions critically influence photostability, yet remain largely unexplored. The inability to resolve defect nature and its evolution at crystallographically distinct grain facets and their boundaries has hindered progress toward industrial-grade stability, as conventional passivation techniques targeting macro-level bulk or surface defects fail to address localized degradation pathways.

Facet orientation engineering has shown promise in mitigating photoinstability arising from misoriented grain facets in polycrystalline perovskite films [18–26]; however, it still faces substantial challenges in achieving precise crystallographic control across perovskites with different compositions and in large-area fabrication. Crucially, most solution-processed perovskite films naturally adopt a polycrystalline structure with mixed-facet orientations, yet paradoxically, their reported photostability varies significantly across studies [21, 24]. This unresolved contradiction underscores an uncharted aspect of facet-interaction physics, where asymmetric carrier recombination associated with defects at hetero-facet grain boundaries being a key driver of instability. To date, no study has specifically identified the types of defects present on different crystallographic facets and at the hidden grain boundaries of polycrystalline perovskite films, nor has it uncovered the mechanisms governing their photostability across distinct crystallographic orientations.

In this work, we establish the first mechanistic link between mixed-facet-induced inter-grain recombination asymmetry and photodegradation in operational perovskite solar cells (PSCs). Through microscale facet-resolved defect characterization and three-dimensional (3D) carrier dynamics mapping, we identify the defect nature at different crystallographical facets of polycrystalline perovskite films. We quantify asymmetrical recombination velocities on the two sides of grain boundaries between typical (100)- and (111)-facet perovskite grains, which are caused by their different defect identities, unveiling how (100)–(111) grain boundary interactions create non-equilibrium hole accumulation at (111)-facet grain

interfaces that accelerate grain degradation under illumination, despite their intrinsic photostability. Instead of facet orientation engineering, we homogenize the inter-grain recombination in mixed-facet PSCs with 3-fluorophenylethylammonium bromide (m-F-PEABr) which specifically targets identified defects at grain boundaries, achieving PSCs with PCEs of near 26.55% and efficiency losses of <10% after 2,200 h under 1-sun full-spectrum illumination.

2 Results and discussion

2.1 Grain facet-dependent defects and photostability

Solution-processed perovskite films typically exhibit mixed (100)–(111) crystallographic orientations across various perovskite compositions, including both iodine-based perovskite compositions and mixed-halide wide-bandgap perovskites [27–30]. Recent studies highlighted facet-dependent moisture stability in PSCs [18], while their photostability linked to facet-specific defects remain inadequately understood. Therefore, we are motivated to first examine the photostability of the mainstream formamidinium lead iodide (FAPbI₃) PSCs with different crystallographic orientations, including (100)-facet-dominant, (100)–(111)-mixed facets, and (111)-facet-dominant perovskites, abbreviated as (100) PVK, (100)–(111) PVK and (111) PVK, which are realized by incorporating different concentration of phenylammonium iodide (PhAI) in the precursor solution (detailed in Methods). The incorporation of bulky PhA⁺ cations into the PbI₂ precursor selectively suppresses growth along the (100) direction, biasing crystallization toward the (111) orientation via preferential FA coordination with Pb-I octahedra, and ultimately resulting in perovskite films enriched in (111) exposed facets, analogous to earlier observations with bulky ammonium salts such as *n*-butylammonium acetate (BAAC) [21]. The orientation-dependent crystallographic structures of these perovskite films are evidenced by X-ray diffraction (XRD) and grazing-incidence wide-angle X-ray scattering (GIWAXS) results shown in Supplementary Fig. 1. We note that the low PhAI content used for the (111) PVK (~0.06 mM mL⁻¹) does not give rise to PhA-derived low-dimensional phases, as confirmed by the lack of low-angle diffraction signatures and negligible AFM-IR responses from PhA⁺ species (Supplementary Fig. 2). Scanning electron microscopy (SEM) results indicate that all films exhibit similar grain sizes, ensuring that the observed differences in photostability are primarily attributed to their crystallographic orientations (Supplementary Fig. 3). As illustrated in Fig. 1a and Supplementary Table 1, the (100)–(111) mixed-facet PSCs undergo the fastest photodegradation, whereas the (111)-facet-dominant PSCs exhibit the highest stability.

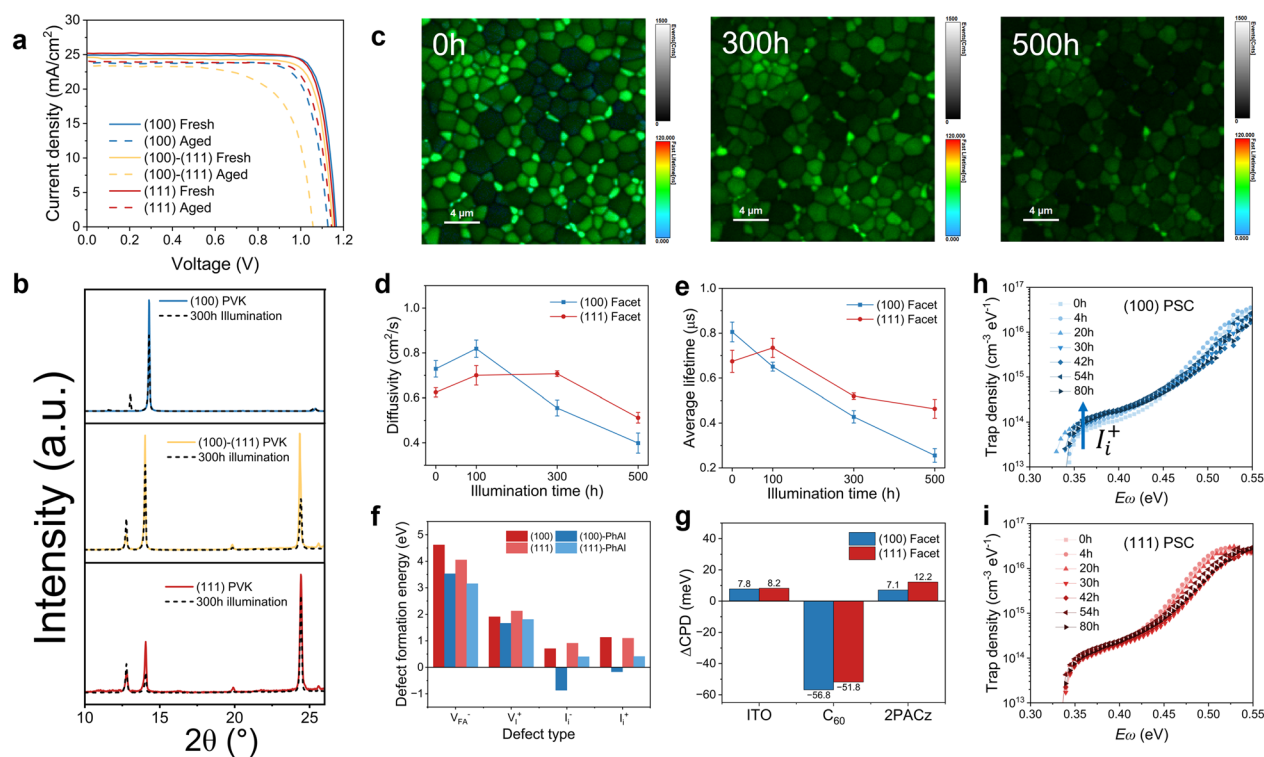


Fig. 1 Grain facet-dependent defect nature and photostability. **a** Current density (J)–voltage (V) curves of fresh and illumination-aged PSCs with (100)-, (100)–(111) and (111)-facet orientations. **b** XRD spectra of fresh and 300 h illumination-aged perovskite film with (100)-, (100)–(111) and (111)-facet orientations. **c** In situ TRPL mapping of (100)–(111) perovskite film under illumination with different time. **d** Dependent of the diffusivity coefficient (D) on illumination time for the (100)- and (111)-facet grains in (100)–(111) perovskite films. **e** Dependent of the average PL lifetime on illumination time for the (100)- and (111)-facet grains in (100)–(111) perovskite films. **f** Formation energies of FA vacancy (V_{FA}^-), iodine vacancy (V_I^+) and iodine interstitial (I_I^- and I_I^+) defects on the (100) and (111) facets of FAPbI₃ perovskite with and without PhAl. **g** The change in contact potential difference before and after illumination (ΔCPD) of (100)- and (111)-facet grains in (100)–(111) perovskite films on ITO, C₆₀/ITO and 2PACz/ITO substrates. **h** tDOS tracking of a (100)-facet-dominated PSC under illumination with different time. **i** tDOS tracking of a (111)-facet-dominated PSC under illumination with different time

To understand the photodegradation behavior of PSCs with different facet orientations, we monitored the changes in XRD peak intensity for the (100) and (111) facets of the films under AM 1.5G 1-sun illumination in an N₂ atmosphere. As shown in Fig. 1b and Supplementary Fig. 4, after illumination, the XRD peak intensity of the (100) facet in (100) PVK dropped significantly, whereas the (111) PVK film showed a much smaller reduction in the peak intensities of the (100) and (111) facets. This result is, to some extent, consistent with previous findings that (111)-facet-dominated film exhibits better photostability than (100)-facet-dominated films on narrow-bandgap and wide-bandgap perovskites [12, 19, 20, 22]. However, surprisingly, for the mixed facets (100)–(111) PVK, the XRD peak intensity of the (111) facet showed a more pronounced decrease than that of the (100) facet. To clearly capture this anomalous degradation behavior, we quantified the relative evolution of the two facets by tracking the (100)/(111) peak-intensity ratio before and after illumination (Supplementary

Fig. 5), which shows that the (100)–(111) PVK exhibits a significantly faster degradation of the (111)-facet grains than either the (100) or (111) PVK, suggesting an unanticipated degradation pathway within the mixed-facet films. In addition, PbI₂ phases are detected in all three types of films, and all samples exhibit an increase in PbI₂ peak intensity upon illumination. Notably, the (100)–(111) PVK films show a PbI₂ evolution behavior comparable to that of the (111) PVK films during photodegradation, despite their clearly distinct facet-dependent degradation characteristics.

To elucidate the above phenomenon, we first investigate the intrinsic photostability of (100) and (111) facets on individual grains from (100)–(111) PVK films using a microscale confocal time-resolved photoluminescence (TRPL) and carrier diffusion mapping system (as schematically shown in Supplementary Information Fig. 8). The laser spot size was kept below 400 nm in diameter, which was smaller than the average grain size of our perovskite films, to avoid the influence of carrier dynamics

from the surrounding grains during measurement. Figure 1c shows the in situ TRPL mapping images of a (100)–(111) PVK film subjected to illumination for different durations. The overall PL intensity of the film gradually decreases as illumination time increases. Then we track the microscale carrier dynamics of individual grains exposing (100) and (111) facets. Previous studies have established that grains exhibiting near-triangular morphologies correspond to exposed (111) facets, whereas grains with more rectangular or non-triangular geometries are associated with (100) facets [18, 20]. To further validate the rigor of this morphology–facet correlation, we fabricated a series of films with systematically varied (111)/(100) facet ratios in term of their XRD peak intensities, ranging from 0.07 to 3.33, and statistically analyzed the dependence of the population of triangular grains on the (111)/(100) ratio. Statistical SEM and PL mapping analyses averaged over ten randomly selected regions of identical area reveal a pronounced increase in the number of triangular grains with increasing (111)/(100) ratio (Supplementary Fig. 6 and 7), corroborating that triangular grains reliably indicate exposed (111) crystal facets. Following this criterion, we tracked the carrier diffusion coefficient (D) and the average PL lifetime for more than 20 individual (100)- and (111)-facet-exposed grains over the course of illumination, as shown in Fig. 1d, e and detailed in Supplementary Information Note 1 and Supplementary Fig. 8–10 [31]. It is found that, on average, (100)-facet-exposed grains exhibit a higher initial D and longer PL lifetimes compared to their (111)-facet-exposed counterparts. However, these values decline more rapidly for (100)-facet-exposed grains under illumination than those for (111)-facet-exposed grains, indicating that the (111) facet itself exhibited a better photostability than the (100) facet.

The above results were obtained from pure films rather than within a device, we infer that the difference in intrinsic facet photostability primarily stems from their different defect nature. To further explore this, we performed density functional theory (DFT) calculations using PbI_2 -terminated perovskite surface models, consistent with the PbI_2 -rich synthesis conditions to compare the formation energies of A-site and halide ion vacancies (V_{FA}^- and V_{I}^+) which were prone to form at the perovskite surfaces, as well as halide ion interstitials (I_i^- and I_i^+) which might also form due to their small size, on both the (100) and (111) facets with and without PhAI decoration [32]. (Supplementary Information Fig. 11–14). Figure 1f presents the corresponding formation energies of V_{FA}^- , V_{I}^+ , I_i^- and I_i^+ on the (100) and (111) facets of FAPbI_3 with and without PhAI on the surface. For both the facets, iodine interstitial defects (I_i^- and I_i^+) are the most likely formed defects given their low formation energies.

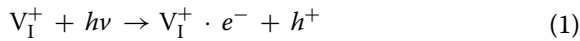
In addition, the (111) facet is likely to be richer in V_{FA}^- defects compared to the (100) facet due to its relatively lower defect formation energy; whereas, the (100) facet may exhibit a higher concentration of V_{I}^+ and I_i^- given their relatively lower formation energy on the (100) facet.

To further understand the charge trapping behaviors of the two facets, we conducted surface photovoltage (SPV) mapping measurements on the (100) and (111) facets of the (100)–(111) PVK film on different substrates, including ITO, fullerene (C_{60})/ITO and 2PACz/ITO. SPV mapping has been routinely adopted to reveal defect-induced charge trapping states in semiconductors by detecting changes in the contact potential difference (CPD) after light excitation [33]. The detailed analysis is provided in Supplementary Note 2 and Supplementary Fig. 15–17. Our findings indicate that the (100)-facet-exposed surface exhibited a greater decrease in CPD after light illumination on an electron extracting substrate C_{60} /ITO than the (111)-facet-exposed surface, where the change in CPD reflected the hole concentration on the surface, suggesting a more hole-charged surface condition for the (100) surface (Fig. 1g). On the contrary, the (111)-facet-exposed surface showed a more pronounced increase in CPD after light illumination on a hole extracting substrate 2PACz/ITO, where the change in CPD is resulted from the surface electron concentrations, indicating that the (111) surface is more electron-rich (Fig. 1g). These results suggest that the (100) and (111) facets are primarily dominated by electron and hole trapping defects, respectively, leading to the accumulation of opposite charge carriers upon illumination [33].

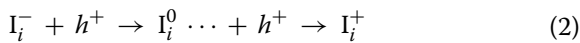
Combining DFT calculations and SPV results, we infer that the main difference in defects distinguishing the (100)- and (111)-facet properties of FAPbI_3 films are V_{I}^+ and V_{FA}^- , respectively. Compensatory I_i^- and I_i^+ defects may still be largely present in both cases, resembling the bulk defect characteristics observed in iodide-based perovskites in our previous studies [10]. The different defect nature exhibited by the two facets leads to distinct defect-induced charge trapping behaviors under illumination, which ultimately influences their intrinsic photostabilities.

To understand how different defect nature influences the photostability of the two facets and the whole film, we measured the trap density of states (tDOS) of perovskite solar cells with (100) PVK and (111) PVK upon illumination with a device structure of ITO/2PACz/perovskite/ C_{60} /BCP/Ag. As shown in Fig. 1h, i, the trap density of the trap band at 0.36 eV in the (100) PVK solar cell increased noticeably after illumination, whereas it remained nearly unchanged in the (111) PVK solar cell. Our previous study identified this trap band as being primarily caused by I_i^+ defects in iodide-based perovskites

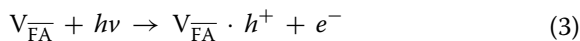
[10]. Therefore, we propose that the defect evolution on the (100) facet under illumination could begin with the trapping of electrons by V_I^+ which causes a hole accumulation:



followed by oxidation reactions, progressing from I_i^- to I_i^0 , and eventually to I_i^+ :



Earlier theoretical calculations have suggested that the transition (2) could occur without significant geometrical rearrangement if sufficient energy was provided, which could be fulfilled by illumination [34]. In contrast, the reverse transition from I_i^+ to I_i^0 involved substantial atomic displacement, making electron detrapping via a non-radiative recombination pathway from I_i^0 more likely. This means that the defect evolution on the (111) surface likely follows a different pathway:



Therefore, the (111)-facet-oriented perovskites demonstrate a better surface photostability compared to their (100)-facet-oriented counterparts.

2.2 Asymmetric inter-grain recombination induced photodegradation

The above results, however, could not well explain the faster photodegradation of (111)-facet grains in the (100)–(111) mixed-facet perovskite observed by XRD. We thus speculate that this unexpected degradation behavior is driven by their hetero-facet grain boundaries rather than grain surfaces. To address this, we employed a self-established technique based on confocal TRPL and carrier diffusion measurements, which enables the evaluation of carrier recombination velocities (S) at hidden grain boundaries within polycrystalline perovskite films [31], providing new insights into the microscale heterogeneity between different-facet grains.

Details of this method have been described in our previously published work [31]. Briefly, S is determined by mapping the transient PL pattern changes near the grain boundary, caused by the defect-induced non-radiative recombination of carriers at the grain boundary (Supplementary Note 3). Here, we select neighboring (100)- and (111)-facet-exposed grains and measure S at their shared grain boundary from both sides, as schematically illustrated in Fig. 2a and exemplified in the TRPL mapping image in Fig. 2b. It is discovered that PL profiles against

the grain boundary vary distinctly over time on the (100)- and (111)-grain sides, despite sharing the same boundary (Fig. 2c). The extracted PL profile peak positions over time (Fig. 2d) reveal that S at the grain boundary of the (100) and (111) grains are around 300 ± 80 and 800 ± 200 cm/s, respectively (Fig. 2e, detailed analysis is provided in Supplementary Note 3 and Supplementary Fig. 18–19). This result suggests an asymmetric carrier recombination at the hetero-facet grain boundary between (100)- and (111)-facet-exposed grains. In contrast, neighboring (100)- or (111)-facet grains exhibit symmetric PL peak position profiles (Fig. 2d), validating that S is similar on both sides of the grain boundary. Statistical results are shown in Supplementary Fig. 20. These findings indicate that for contacting (100)- and (111)-facet-exposed grains, the (111)-facet grains are more defective at the grain boundary, highlighting distinct defect characteristics on the two sides of their grain boundaries.

To further understand the impact of asymmetric grain boundary recombination on carrier dynamics in perovskite films, we measured the overall TRPL of (100) PVK, (100)–(111) PVK and (111) PVK films. As shown in Fig. 2f, the (100) PVK film exhibited the longest PL lifetime, which aligns with its longer initial PL lifetime measured on single grains (Fig. 1e) and smaller S at grain boundaries. Meanwhile, the (111) PVK film displayed a relatively shorter PL lifetime, consistent with its shorter initial PL lifetime on the (111) surface (Fig. 1e) and larger S at grain boundaries. Notably, the (100)–(111) PVK film exhibited the fastest PL decay among the three films, suggesting that the asymmetric defect nature of the grain boundary enhances carrier recombination within the film through inter-grain carrier recombination.

Then we carried out positron annihilation spectroscopy (PAS) measurements on the three types of perovskite films, which mainly detects neutral or negatively charged vacancy-defects in perovskites, i.e., Pb vacancies (V_{Pb}^{2-}) and V_{FA}^- . Figure 2g displays the positron annihilation lifetime spectra of (100) PVK, (100)–(111) PVK and (111) PVK films, respectively. A magnified view of the channel range from 1030 to 1070 (Fig. 2g) reveals that the three spectra are not fully coincident, resulting in discernible differences in the positron lifetime values extracted via PATFIT analysis. The dominant positron lifetime extracted from the spectra using a three-component fitting method was 387.4 ± 11.9 ps, 387.6 ± 9.6 ps and 404.4 ± 12.1 ps for the (100) PVK, (111) PVK and (100)–(111) PVK, respectively (Supplementary Information Table 2). Although it is difficult to quantify the contributions of V_{Pb}^{2-} or V_{FA}^- to the overall positron lifetime, the observed increase in lifetime from ~ 387.4 ps to 404.4 ps in FAPbI₃ follows a trend akin to that found in MAPbI₃ and MAPbBr₃ where V_{Pb}^{2-} and V_{MA}^- contribute

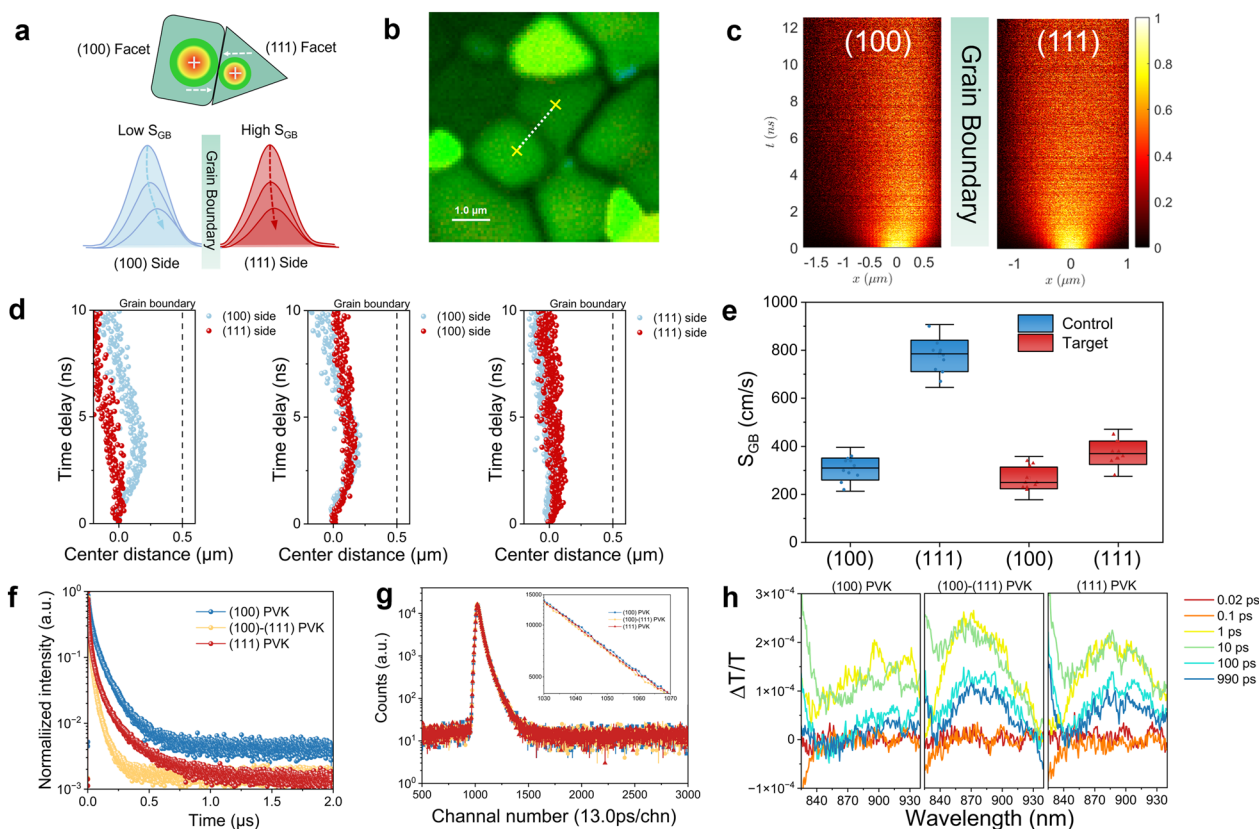


Fig. 2 Defect-dependent asymmetric inter-grain recombination. **a** Schematic of carrier diffusion imaging to measure grain boundary recombination velocities (S_{GB}) at the (100)- and (111)-facet grain boundary. A small S_{GB} value causes the PL profile center shifts toward the grain boundary, and a large S_{GB} value causes the PL profile center shifts against the grain boundary. **b** PL mapping image of a (100)–(111) perovskite film. The rectangular and triangular grains are (100)- and (111)-facet-exposed grains, respectively. The yellow crosses denote the laser spot sites for the measurement of S_{GB} on the two grains. The dotted line indicates the linear profile direction for the time-dependent carrier diffusion mapping as shown in **c**. **c** Measured carrier diffusion profiles on the two sides of (100)- and (111)-facet adjacent grain boundary. **d** Extracted carrier diffusion profile center (peak intensity) from **c** of (100)–(111) grain boundary, and together with those of (100)–(100) and (111)–(111) adjacent grain boundaries in (100)–(111) perovskite film. **e** Statistic results of S_{GB} of (100)- and (111)-facet grains in control and target perovskite films. **f** TRPL decay of (100) PVK, (100)–(111) PVK and (111) PVK. **g** Positron annihilation lifetime spectra of (100) PVK, (100)–(111) PVK and (111) PVK. The inset displays a magnified view of the spectral region corresponding to channel numbers 1030–1070. **h** Transient absorption spectra of (100) PVK, (100)–(111) PVK and (111) PVK.

to positron lifetimes of approximately 350–377 ps and 401–442 ps, respectively [35, 36]. Notably, the presence of larger A-site cation vacancies correlates with significantly longer positron lifetimes [36]. These results suggest that V_{FA}^- is more prevalent in (100)–(111) PVK compared to both (100) PVK and (111) PVK films. The high concentration of V_{FA}^- is believed to be located at the grain boundary toward the (111)-facet-exposed grain side, contributing to the high defect density measured by confocal TRPL mapping. While for the (100)-facet grain, since most grain boundaries in the (100)–(111) PVK film are nearly *vertically* oriented (Supplementary Information Fig. 21), the grain boundary of the (100)-facet grain is expected to exhibit a similar defect nature with the (100) surface.

To further validate the distinct defect characteristics of the (100)- and (111)-facet grains and their influence on carrier dynamics under illumination, we conducted transient absorption (TA) measurements on the (100) PVK, (111) PVK, and (100)–(111) PVK samples. As shown in Supplementary Information Fig. 22 and Fig. 2h, in addition to the ground-state bleaching signal observed at 780 nm and photo-induced absorption at 810 nm, all samples exhibited positive $\Delta T/T$ signals in the sub-bandgap range of 840–940 nm within a time window of 0.1–990 ps, which are attributed to defect-state-induced absorption bleaching [37]. For the (100) PVK, the sub-bandgap $\Delta T/T$ peaked around 920 nm and decayed rapidly within 100 ps. In contrast, the (111) PVK showed a sub-bandgap peak near 880 nm

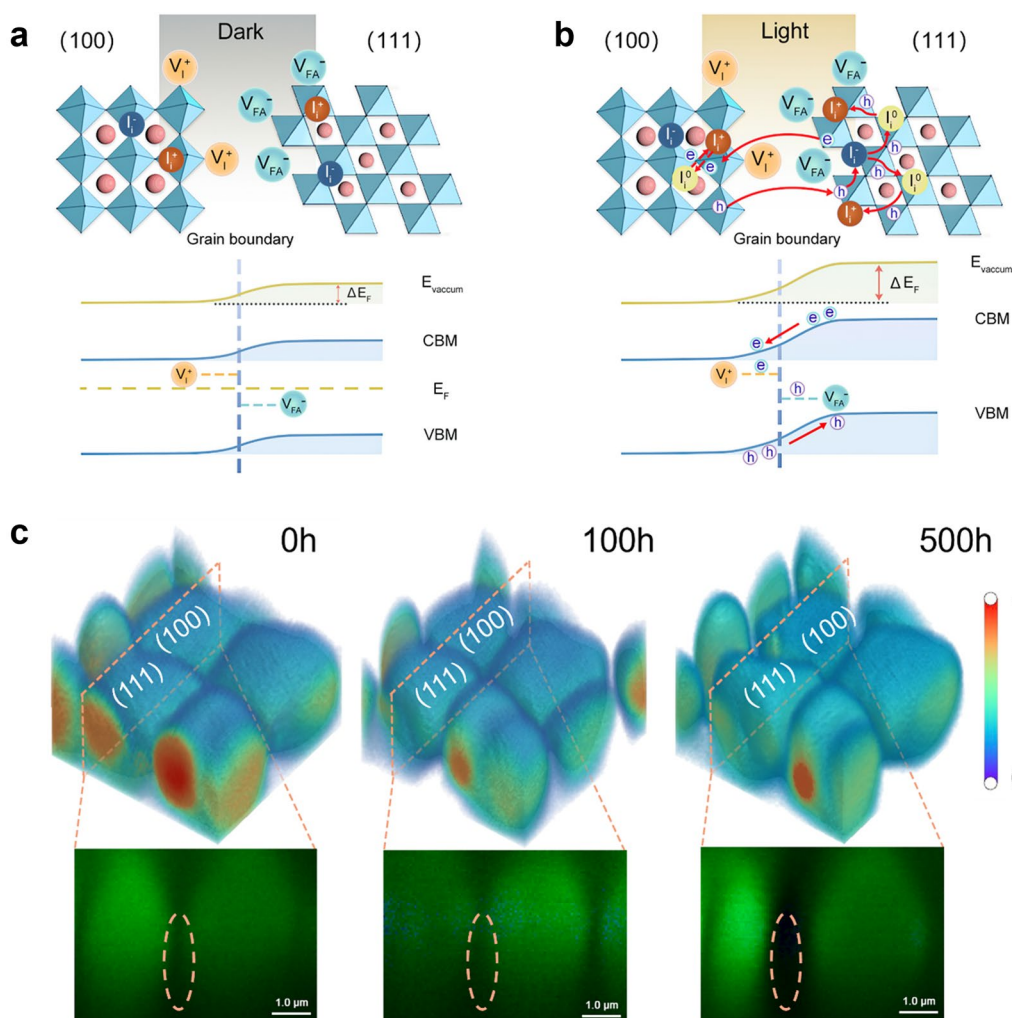


Fig. 3 Asymmetric inter-grain recombination induced photo instability. Schematic of ion dynamics between the (100)- and (111)-facet-exposed grains in mixed-facet perovskite films **a** in the dark and **b** under illumination. The dominant defects of V_I^+ (orange circle) and V_{FA}^- (cyan circle) on the (100) and (111) grain boundaries form a microscale electric field pointed from the (100)-facet grain to the (111)-facet grain in the dark. This electric field is enlarged upon illumination when photo-generated holes and electrons are initially accumulated on the (100)- and (111)-facet grain boundaries, respectively, and drives charge transfer between the two grains. After inter-grain charge transfer, I_i^- was transformed into I_i^0 and I_i^+ with involvement of transferred holes within the (111)-facet grains, dragging the degradation of (111)-facet grains at the grain boundaries. **c** High-resolution 3D PL mapping reconstruction of (100)-(111) perovskite grains before and after illumination with 100 and 500 h. The (111)- and (100)-facet grains are marked on the 3D PL mapping diagram. The color bar indicates the normalized PL intensity. The orange dashed rectangles indicate the locations of the cross-sectional PL mapping obtained across the (111)- and (100)-facet grains. The dashed ellipses denote the grain boundary area between the (111)-facet grain (left) and the (100)-facet grain (right)

with significantly longer transition lifetimes extending beyond 990 ps. We infer that this is reflected by the different dominant defect of V_I^+ and V_{FA}^- on the (100)- and (111)-facet grains, respectively. Notably, the (100)-(111) PVK displayed a transient sub-bandgap signal similar to that of the (111) PVK but with even higher intensity, further confirming a higher concentration of V_{FA}^- defects at the grain boundaries, considering the relatively minor contribution from the surface (111)-facet grains.

In this scenario, the grain boundary between (100)- and (111)-facet-exposed grains is enriched with hole-trapping defects V_{FA}^- on the (111) grain side and electron trapping defects V_I^+ on the (100) grain side, respectively, thus forming a microscale inter-grain electric field directed from the (100)-facet grain toward the (111)-facet grain, as schematically illustrated in Fig. 3a. Upon light illumination, photo-excited electrons and holes are trapped and recombine at the (100)-facet grain boundary side and (111)-facet grain boundary side, respectively, resulting

in surpluses of holes and electrons on opposite sides of the hetero-facet grain boundary (Fig. 3b). This charge imbalance further enhances the band bending between the two grains, and consequently, holes are driven from the (100)-facet grain to the (111)-facet grain, while electrons migrate in the opposite direction. This charge transfer leads to hole accumulation at the (111)-facet grain boundary, initiating photodegradation as described in reaction (2). Additionally, the inter-grain transported carriers could recombine with residual carriers left by surface-trapped charges on the (111)- and (100)-facet grains, respectively. Such inter-grain recombination ultimately contributes to the performance degradation of PSCs.

To validate the hetero-facet grain boundary driven photodegradation for the (111)-facet-exposed grains in mixed-facets films, we reconstructed high-resolution 3D PL intensity mapping of the (100)–(111) PVK film before and after illumination to visualize the degradation pathway at sub-micron scale hidden grain boundaries (Fig. 3c). The reconstruction procedure is described in Supplementary Note 4 and Supplementary Fig. 23. This technique enables the non-destructive investigation of carrier recombination near buried grain boundaries while minimizing interference from surface PL signals. After illumination, the (111)-facet grain exhibited significantly accelerated degradation at grain boundaries compared to both the adjacent (100)-facet grains and their own grain surfaces. These results provide direct evidence of a degradation pathway originating from the grain boundary in (111)-facet grains within mixed-facet films, underscoring the need to mitigate imbalanced inter-grain recombination to enhance solar cell photostability.

Crucially, the coupling between this non-equilibrium charge dynamics and the subsequent structural evolution of the grain boundaries represents a profound, yet fundamentally underexplored, degradation pathway. While classical grain boundary migration or rotation typically observed in hard metallurgy might be restricted in soft hybrid lattices, the severe hole accumulation at the (111)-facet boundaries undoubtedly triggers localized structural destabilization. Specifically, the deep oxidation of interstitial iodide driven by localized holes disrupts the coordination of the $[\text{PbI}_6]^{4-}$ octahedral framework. The ensuing electrostatic stress naturally drives localized amorphization or phase transformation, precipitating the irreversible formation of PbI_2 phases strictly along these hetero-facet boundaries. At present, the extreme electron-beam sensitivity of hybrid perovskites precludes the direct real-time visualization of these operando microstructural dynamics via traditional high-resolution transmission electron microscopy. Unmasking this charge-coupled structural evolution remains a critical

frontier. We anticipate that future investigations employing operando synchrotron-based scanning X-ray diffraction microscopy (nano-XRD), ultra-low-dose in situ environmental TEM, or macroscopic phase-field modeling integrated with non-adiabatic molecular dynamics (NAMD) will be essential to fully decouple the dynamic interplay between localized charge accumulation and the ultimate structural collapse of grain boundaries.

To further elucidate whether, and to what extent, facet mixing drives asymmetric inter-grain degradation that preferentially accelerates the decay of (111)-facet grains, we investigated the facet-resolved degradation behavior of perovskite films with varied (111)/(100) facet ratios, spanning a wide range from 0.07 to 3.33 as quantified from relative XRD peak intensities (Supplementary Fig. 24). As shown in Supplementary Fig. 24, the films exhibit a clear and non-monotonic dependence of photostability on the (111)/(100) facet ratio. When the (111)/(100) ratio falls within the range of 0.77–2.25, the films undergo progressively faster photodegradation after 300 h of continuous illumination. In contrast, when the films become (111)-facet-dominant with a (111)/(100) ratio of 3.33, their photostability is markedly restored. These results undoubtedly demonstrate that mixed-facet films exhibit inferior photostability compared with either (100)- or (111)-facet-dominant counterparts where the (111)-facet grain degraded even faster.

2.3 Homogenizing inter-grain recombination for photostable PSCs

To eliminate asymmetric inter-grain recombination, homogenizing the defect landscape between the (100) and (111) grains are crucial. The selection of m-F-PEABr is driven by its synergistic molecular architecture for deep defect remediation. Due to its bulky backbone, m-F-PEABr is sterically excluded from the 3D perovskite lattice, selectively segregating into the highly defective grain boundaries (GBs). Here, its compact fluorinated benzene ring and flexible ethylammonium chain provide precise nanoscale accessibility to penetrate the confined topological spaces of (111)-adjacent boundaries. Once localized at these V_{FA}^- -enriched boundaries, the molecule executes a synergistic dual-action passivation mechanism. Primarily, the terminal ammonium group ($-\text{NH}_3^+$) mimics the missing FA^+ cation, physically filling the vacancy to neutralize the local charge imbalance via robust hydrogen bonding. Crucially, this primary coordination is firmly locked by the meta-fluorinated benzene ring. The extreme electronegativity of the meta-fluorine generates a strong local dipole, geometrically perfectly angled to engage in weak $\text{C}-\text{F}\cdots\text{I}$ coupling with exposed iodide ions on the (111) facet [38, 39]. Unlike ortho- or para-substitutions that might induce steric repulsion,

the meta-isomer provides the optimal spatial distance for this secondary stabilization, as confirmed by our XPS results (Supplementary Fig. 26).

Importantly, this dual-anchoring mechanism unveils a universal selectivity principle for rational passivator design. This defect-targeting strategy can be broadly generalized to other fluorinated aromatic ammonium salts. However, their efficacy will be stringently dictated by isomer configurations and the number of fluorine substitutions, as the specific atomic arrangement of the (111) facet imposes strict spatial requirements. Future molecular designs must ensure these secondary dipole interactions precisely align with the lattice without perturbing pristine structural integrity.

Here, we refer to the perovskite films and solar cells without and with the m-F-PEABr additive as control and target samples, respectively. Please note that both the control and target films are additionally surface-passivated by piperazine hydroiodide (PipI) to minimize the surface recombination loss gap, leveraging its proven effectiveness in passivating surface defects in PSCs [40].

To validate the grain boundary-selective distribution of m-F-PEABr at the atomic scale, we employed atomic force microscopy infrared spectroscopy (AFM-IR) mapping, which integrates nanomechanical topography with chemical fingerprinting. As shown in Fig. 4a, b, the characteristic m-F-PEA⁺ signal at 1243 cm⁻¹ which represent the C–F vibration [41, 42] (Supplementary Fig. 27) is intensely localized at grain boundaries, with negligible presence within grain interiors. To further verify whether the additive m-F-PEA⁺ is predominantly located at grain boundaries rather than on the film surface, we examined the F 1s XPS spectra and IR-mapping signals of m-F-PEA⁺ in films prepared with m-F-PEABr either as a bulk additive or as a surface passivator, both before and after IPA rinsing. Notably, the IPA rinsing significantly reduces the F 1s XPS and IR signals when m-F-PEABr is applied as a surface passivator, whereas these signals remain essentially unchanged when m-F-PEABr is incorporated as an additive (Supplementary Fig. 28–29). This spatial specificity confirms that m-F-PEABr preferentially segregates to grain boundary regions rather than adsorbing

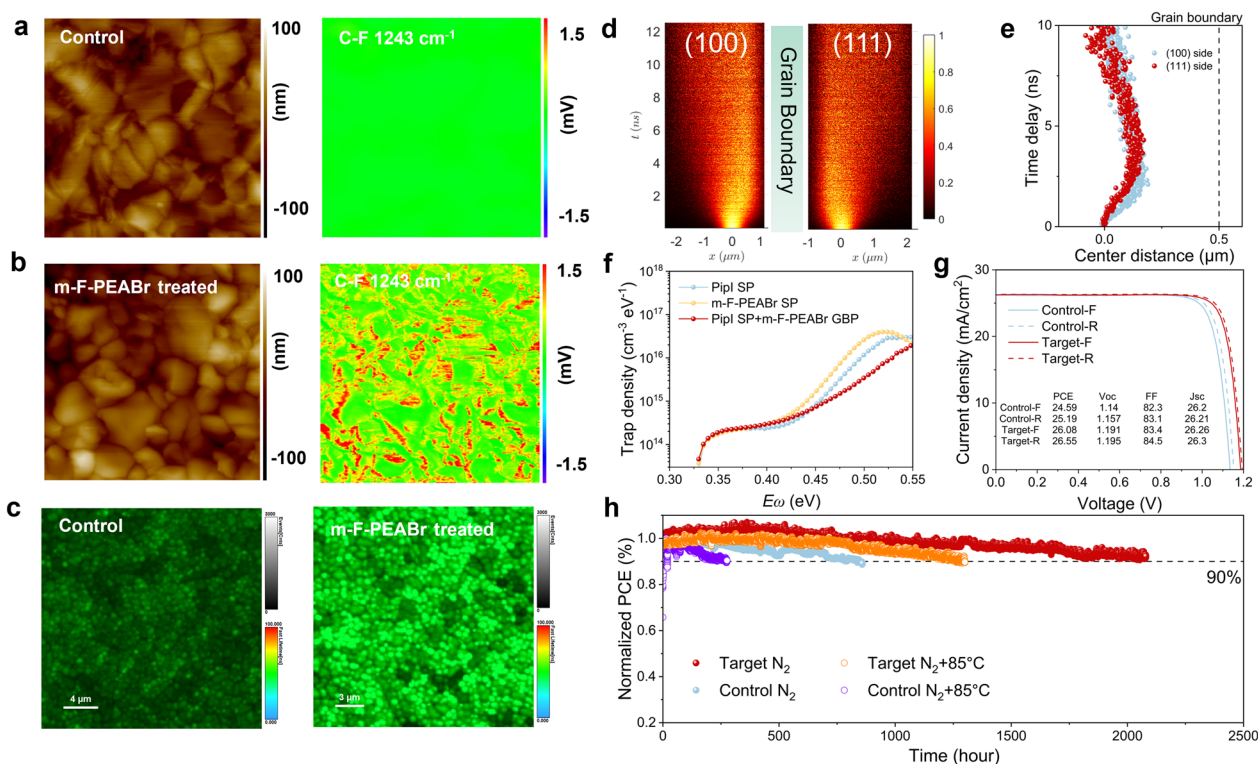


Fig. 4 Homogenizing inter-grain recombination for photostable PSCs. AFM topography (left) and IR images (right) of **a** control and **b** m-F-PEABr treated (100)–(111) PVK films. The IR signal was measured at 1243 cm⁻¹ which corresponding to the C–F vibration in m-F-PEA⁺. **c** TRPL mappings of control and m-F-PEABr treated (100)–(111) PVK films. **d** Measured carrier diffusion profile near the (100)- and (111)-facet adjacent grain boundary of (100)–(111) PVK films after m-F-PEABr treating. **e** Carrier diffusion profile center tracking of (100)–(111) adjacent grain boundary on the (100)-facet side (blue dots) and the (111)-facet side (red dots). **f** tDOS spectra of (100)–(111) PSCs with PipI surface passivation (SP), m-F-PEABr SP, and PipI SP + m-F-PEABr grain boundary passivation (GBP). **g** J–V curves of control and target PSCs. **h** MPP stability tracking of unencapsulated control and target devices under 1-sun illumination at room temperature and 85 °C in N₂ atmosphere

onto facet surfaces. TRPL mapping further indicates that the incorporation of m-F-PEABr significantly enhances the PL intensity within individual grains of the polycrystalline film (Fig. 4c), indicating effective suppression of recombination at grain boundaries.

Most importantly, we quantified the carrier recombination at the grain boundaries of (100)- and (111)-facet grains in the target film, and found that the target film presented a significantly improved balance in inter-grain recombination (Fig. 4d, e), with the S at the grain boundary on the (100)- and (111)-facet grain sides being approximately 300 ± 80 and 320 ± 100 cm/s, respectively (Fig. 2e). The tDOS result shows an obviously decrease of trap density in the target PSC, which was not observed in devices with m-F-PEABr as a surface passivator (Fig. 4f). This indicates the effective passivation of grain boundary defects by m-F-PEABr in the target device.

Steady-state PL and TRPL further revealed a significant reduction in non-radiative recombination loss and a substantial increase in carrier lifetime in perovskite films upon incorporating m-F-PEABr (Supplementary Fig. 30). Consequently, the target PSCs demonstrated a notable enhancement in device performance compared to the control devices, with the PCE of the champion devices increasing from 25.19 to 26.55% and a featured open-circuit voltage (V_{OC}) of 1.195 V, fill factor (FF) of 84.5% and short-circuit current density (J_{SC}) of 26.30 mA/cm^2 (Fig. 4g). Statistical device data (Supplementary Fig. 31) further confirm an overall performance improvement upon m-F-PEABr treatment, with the average V_{OC} increased by 0.035 V, the average fill factor improved by 2%, and the average J_{SC} enhanced by 0.2 mA cm^{-2} . External quantum efficiency (EQE) measurements yield integrated current densities of 25.62 mA cm^{-2} for the target champion device, in good agreement with the measured J_{SC} values with a deviation of less than 3% (Supplementary Fig. 32).

To evaluate the photostability of the devices, an operational stability test for both the control and target PSCs following the ISOS-L-1 and ISOS-L-2 testing protocol, in which unencapsulated devices were exposed to continuous 1-sun equivalent illumination at ambient and 85°C , respectively, from white light-emitting diode (LED) (Supplementary Fig. 33) and maintained at the maximum power point (MPP) voltage. The m-F-PEABr-treated PSCs achieved a T90 (the time it takes for the device's PCE to drop to 90% of its initial value) of 2200 h after continuous MPP tracking at room temperature in an N_2 -filled container, whereas the control devices dropped 10% of its initial efficiency after only 800 h (Fig. 4h). Furthermore, the m-F-PEABr-treated PSCs achieved a T90 of 1250 h after continuous MPP tracking at 85°C (Fig. 4h), significantly outperforming the control devices,

which degraded within just 270 h. These results demonstrate the remarkable photo-thermal stability conferred by the incorporation of m-F-PEABr, highlighting the importance of controlling microscale inter-grain charge recombination for the further improvement of the photostability of PSCs. While the m-F-PEABr-treated devices exhibit remarkable operational stability at 85°C , primarily because the synergistic dual-anchoring (V_{FA}^- filling and C-F...I coupling) kinetically frustrates passivator desorption, assessing the ultimate thermal limits requires further consideration. The inherent acidity of primary ammonium cations fundamentally constrains extreme prolonged high-temperature ($>120^\circ\text{C}$) operation, where excessive thermal energy could eventually trigger unavoidable deprotonation ($\text{RNH}_3^+ \rightleftharpoons \text{RNH}_2 + \text{H}^+$) and subsequent lattice collapse. To surpass this theoretical boundary, future molecular designs should evolve toward resonance-stabilized cations, such as guanidinium or amidinium. By extensively delocalizing the positive charge, these highly symmetric functional groups drastically lower their intrinsic acidity. Integrating such robust terminals with tailored fluorinated aromatic backbones could effectively decouple targeted topological passivation from thermal deprotonation risks, providing a blueprint for ultra-stable perovskite photovoltaics.

3 Conclusion

In conclusion, we unveil a significant photodegradation pathway in polycrystalline PSCs with mixed facets, governed by asymmetric inter-grain carrier recombination due to varying defect natures across grain facets. Through microscopic defect and carrier dynamic characterizations, we have identified the primary defects that distinctly influence the photostability of (100) and (111) facets of FAPbI_3 perovskite, which are V_I^+ and V_{FA}^- , respectively. The different defect nature on the two sides of attached (100)- and (111)-grain boundaries leads to non-equilibrium hole accumulation at (111)-facet grain side, degrading grains despite their inherent photostability. By mitigating imbalanced inter-grain recombination with the incorporation of the selected grain-boundary passivator m-F-PEABr, we have significantly increased the PCE performance of 26.55% and enhanced the photostability of PSCs, achieving T90 operational stability exceeding 2200 h.

4 Methods

4.1 Materials

Materials used in this experiment include lead (II) iodide (PbI_2 , 99.99%, Sigma-Aldrich), formamidinium iodide (FAI, Greatcell Solar Materials), methylammonium iodide (MAI, 99.5%, Greatcell Solar Materials), methylamine hydrochloride (MACl , $>99.8\%$, Xi'an Yuri Solar

Co., Ltd.), chlorobenzene (CB, 99.8%, Sigma-Aldrich), isopropanol (IPA, 99.8%, Sigma-Aldrich), *N,N*-dimethylformamide (DMF, 99.8%, Sigma-Aldrich), dimethylsulfoxide (DMSO, 99.9%, Sigma-Aldrich), bathocuproine (BCP, Sigma-Aldrich), fullerene (C_{60} , 99%, Sigma-Aldrich), [2-(9H-Carbazol-9-yl)ethyl]phosphonic acid (2PACz, 99.5%, TCI) Piperazinehydroiodide (PipI, 99.5%, Xi'an Yuri Solar Co., Ltd.), phenylammonium iodide (PhAI, 99.5%, Xi'an Yuri Solar Co., Ltd.), nickel oxide (NiO_x , 99.5%, Xi'an Yuri Solar Co., Ltd.), 3-fluorophenylethylammonium bromide (m-F-PEABr, 99.5%, Xi'an Yuri Solar Co., Ltd.). All materials were used as received without any purification.

4.2 Preparation of perovskite films

(100) PVK, (100)–(111) PVK and (111) PVK: perovskite films with different facet orientations were fabricated using a two-step sequential deposition method. The perovskite film composition used in this study is $FA_{0.93}MA_{0.07}PbI_3$. For the PbI_2 precursor solution, PbI_2 (1.5 M) was dissolved in DMF/DMSO (V:V = 9:1), which was stirred at 65 °C for 2 h before use. The solution of FAI for the second step was stirred at room temperature for 2 h. MAI and MACl were incorporated as additives with a weight ratio of FAI:MAI:MACl = 90 mg:6 mg:9 mg in 1 mL IPA. PhAI (optimized molar ratio of 0%, 1%, 4% relative to PbI_2) was added in PbI_2 precursor solution to prepare (100) PVK, (100)–(111) PVK and (111) PVK, respectively. Bare glasses were cleaned sequentially by aqueous detergent, de-ionized water, acetone and ethanol for 20 min each step, then treated by ultraviolet (UV)-ozone for 15 min before use. PbI_2 precursor solutions were spin-coated at 1500 r.p.m. for 30 s, then annealed at 70 °C for 1 min in glove box. Next, FAI precursor solution was spin-coated on PbI_2 films at 1800 r.p.m. for 30 s in a glove box, which were then annealed at 150 °C for 15 min in ambient environment with 20–30% relative humidity.

4.3 Device fabrication

Patterned indium tin oxide (ITO) glasses were cleaned sequentially by aqueous detergent, de-ionized water, acetone and ethanol for 20 min each step, then treated by UV-ozone for 15 min before use. The nickel oxide (NiO_x) precursor solution (10 mg/ml in de-ionized water) was spin-casted onto ITO glasses at 4000 r.p.m. for 30 s, followed by annealed at 120 °C for 30 min under ambient atmosphere. Then, the substrate was transferred to a glove box filled with N_2 atmosphere. 2PACz (0.5 mg/ml in IPA) was spin-coated on NiO_x /ITO substrate at 4000 r.p.m. for 30 s and annealed at 105 °C for 10 min. After spin-coating the aforementioned perovskite layer, the film was transferred to a nitrogen-filled glovebox for surface passivation. PipI solution (0.3mg/ml in IPA)

was spin-coated on the surface of perovskite film and annealed at 105 °C for 5 min. Finally, transfer the above-mentioned substrate into a vacuum chamber and thermally evaporate C_{60} (20 nm), BCP (6 nm), and silver (Ag, 80 nm) sequentially.

4.4 Structural characterization

X-ray diffraction (XRD) was measured by a Bruker D8 Advance diffractometer with a Cu $K\alpha$ radiation ($\lambda = 1.5406$ Å). The scanning rate was 4°/min at the step-scan size of 0.02°. The perovskite samples for XRD were prepared on glass substrates. Morphology of the samples were measured by a JEOL JSM-7600F scanning electron microscope (SEM). The perovskite samples for SEM measurement were fabricated on ITO substrates. The surface of the samples was coated with platinum (Pt) using a sputter to prevent the charging during the measurement. Grazing-incidence wide-angle X-ray scattering (GIWAXS) measurement were accomplished with a Xeuss 2.0 WAXS/SAXS laboratory beamline using a Cu X-ray source (8.05 keV, 1.54 Å) and a Pilatus3R 300K detector. GIWAXS samples were prepared on glass substrate by spin coating.

4.5 TRPL mapping and carrier diffusion coefficient measurement

TRPL mapping was measured by using a PicoQuant Micro Time 100 confocal microscopic system. An Olympus 100×oil immersion objective with a numerical aperture of 1.45 was used for the measurement. Samples were covered by immersion oil (Olympus Immersion Oil Type-F) in a dark environment during the measurement. Picosecond 640 nm pulsed laser with a repetition rate of 5 MHz was used to measure the TRPL mapping of perovskite films with composition of $FAPbI_3$. The average laser intensity was kept at around 12 nW at the repetition rate of 5 MHz, corresponding to a photon energy of ~2.4 eV per pulse. The PL generated by samples was collected through the same objective and relayed onto an image plane through a telescopic relay lens system. The transmitted epifluorescence image was then collected by a single photon detector (PicoQuant PMA Hybrid) through an optical fiber that was connected to a precise x–y piezo motor stage (PI U-751.24) with a linear step resolution of 100 nm. Carrier diffusion coefficient (*D*) was measured by using the PicoQuant Micro Time 100 confocal microscopic system. The 640-nm laser was positioned at the center of the target grains and the excited PL signal was collected via a movable optical fiber to the detector. The minimum temporal resolution of the system was 25 ps, which was determined by the time-correlated single-photon counting modules (TimeHarp 260).

4.6 Steady-state PL and TRPL measurement

Steady-state PL was performed on samples in air using a Lumin QY manufactured by Shanghai Ideaoptics Co., LTD. A 515 nm laser was used as the excitation source. TRPL spectra of micro-grain and perovskite film was performed using the PicoQuant Micro Time 100 confocal microscopic system with 100×oil length and 10×length, respectively. The integration time for each point was 120 s. The repetition rate of 640 nm laser was set at 0.5 MHz to ensure that the complete decay of the fluorescence signal.

4.7 Positron annihilation spectroscopy (PAS)

PAS was performed using a conventional fast-fast coincidence spectrometer. A ^{22}Na source (activity $\approx 12 \mu\text{Ci}$) was sandwiched between two identically sized and thick (sample thickness $\approx 1\text{--}2 \text{ mm}$) sheet-like samples, wrapped in tin foil, and positioned between two detectors for measurement. Each spectrum contained 10^6 count points, and positron lifetime parameters were extracted via deconvolution using PATFIT software as following:

$$N_t = \frac{\sum iI_i}{\tau_i} \exp\left(-\frac{t}{\tau_i}\right), \quad (5)$$

whereas τ_i are the fitting relaxation time with different positively charged defects and I_i are the τ_i relative intensities.

4.8 Surface photovoltage measurements (SPV)

The SPV measurements were performed using Bruker Dimension Icon Scanning Probe Microscopy with Nanoscope V controller. All the KPFM images were collected in lift mode by using SCM-PIT-V2 probes (Pt/Ir-coated probes) with a spring constant of $\sim 3 \text{ N/m}$ and a resonant frequency of $\sim 75 \text{ kHz}$. The excitation source is a 405-nm laser and during all KPFM measurements, the optimal lift height is 100 nm.

4.9 Transient absorption (TA) spectroscopy

For femtosecond transient absorption spectroscopy, the fundamental output from Yb:KGW laser (1030 nm, 220 fs Gaussian fit, 100 kHz, Light Conversion Ltd) was separated to two light beam. One was focused onto a BBO crystal to generate 515 nm pulse for pump beam, the other was focused onto a YAG crystal to generate continuum white light as probe beam. The temporal delay between them is controlled via a motorized delay stage. The pump and probe overlapped on the sample at a small angle less than 10° . The transmitted probe light from

sample was collected by a linear array detector (TA-100, Time-Tech Spectra, LLC).

4.10 Atomic force microscopy-based infrared spectroscopy (AFM-IR)

The AFM-nano-infrared examinations were performed with a Bruker Anasys nano IR2-fs instrument. The testing wavelength number was set at 1243 cm^{-1} to detect C–F vibration and 846 cm^{-1} to detect PhA^+ group.

4.11 Thermal admittance spectroscopy (TAS) measurements

Thermal admittance spectroscopy (TAS) measurements: the capacitance–frequency (C–f) spectra were measured by a LCR meter (Keysight E4980A) at frequencies from 0.02 to 2000 kHz with an alternating voltage (20 mV). The C–f spectra were differentiated to calculate the trap density of state (tDOS) versus the energy level (E_ω) of the trap state via the equations:

$$E_\omega = kT \ln\left(\frac{\omega_0}{\omega}\right), \quad (6)$$

$$N_t = -\frac{V_{bi}}{qAW} \frac{dc}{d\omega} \frac{\omega}{kT}, \quad (7)$$

where ω is the angular frequency, ω_0 is attempt-to-escape frequency which was adopted from our previous work calculated on a similar perovskite composition, k is Boltzmann constant, T is the temperature, V_{bi} is the built-in potential, W is the width of depletion region and A is the active area of the device.

4.12 J–V measurement methodology

The J–V characteristics of the photovoltaic devices were measured using a Keithley 2400 source-measure unit under simulated AM 1.5G illumination provided by an Oriel 9600 solar simulator at an intensity of 100 mW cm^{-2} . The light intensity was calibrated using a National Institute of Standards and Technology (NIST)-certified monocrystalline silicon reference cell (Newport 532, ISO 1599). Small-area devices with an active area of approximately 0.06 cm^2 were measured using a shadow mask with a rectangular aperture of $2.5 \text{ mm} \times 2.4 \text{ mm}$. J–V curves were recorded under ambient conditions using both reverse and forward voltage scans. The reverse scan was performed from 1.25 to -0.1 V with a step voltage of -0.02 V and a delay time of 0.2 s per step, while the forward scan was conducted from -0.1 to 1.25 V with a step voltage of 0.02 V and a delay time of 0.2 s per step.

4.13 Stability test

The operational stability of the devices was measured under continuous white light illumination from

a light-emitting diode (LED) with an integrated power density calibrated to 100 mW cm^{-2} . The exact emission spectrum of the LED used for aging is provided in Supplementary Fig. 33. The unencapsulated devices were kept in an N_2 -filled container at ambient temperature and 85°C and maintained at the maximum power point (MPP) voltage.

4.14 DFT calculation

All the density functional theory (DFT) calculations were performed by Vienna ab initio simulation package (VASP). The exchange–correlation interactions were treated with the generalized gradient approximation (GGA) parameterized by the Perdew–Burke–Ernzerhof (PBE) functional. Ultrasoft pseudopotentials were employed to describe electron–ion interactions, with a plane-wave cutoff energy of 700 eV to ensure convergence of the electronic wavefunctions. A Monkhorst–Pack k-point grid of $4 \times 4 \times 1$ was adopted for Brillouin zone sampling, balancing computational accuracy and efficiency for surface systems.

To model surface systems, a supercell approach was adopted with a vacuum layer thickness of $20 \text{ \AA}$ along the non-periodic direction to eliminate spurious interactions. Four types of defective FAPbI_3 surface slabs were constructed: an iodine vacancy V_I^+ , an FA vacancy V_FA^- , and two distinct iodine interstitial defects I_I^+ and I_I^- . Geometry optimizations were performed using the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm combined with density mixing. Long-range van der Waals interactions were incorporated via the DFT-D3 dispersion correction scheme.

Supplementary Information

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Supplementary material 1

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Author contributions

G.Y., Z.N. and D.Y. supervised the project. Z.N. and H.Z. designed the experiments. H.Z., B.Y. and D.Z. fabricated perovskite solar cells. H.Z. carried out the PL mapping and carrier diffusion measurements. Y.G. and H.X. carried out the capacitance measurements for the devices. F.Z. performed DFT calculation and 3D PL mapping reconstruction. P.C., H.Z. and X.L. carried out the GIWAXS measurement. F.Z. and J.J. carried out the SPV measurements. Y.Z., R.Z. and X.M. contributed to the crystallographic measurement and analysis. H.Z., P.H. and Z.N. wrote the paper, and all authors reviewed the paper.

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Data availability

No datasets were generated or analyzed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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References

- Yun Y et al (2025) Dimensional engineering of interlayer for efficient large-area perovskite solar cells with high stability under ISOS-L-3 aging test. *Sci Adv* 11:eadp3112
- Jiang Q et al (2022) Compositional texture engineering for highly stable wide-bandgap perovskite solar cells. *Science* 378:1295–1300
- Zhu H et al (2023) Long-term operating stability in perovskite photovoltaics. *Nat Rev Mater* 8:569–586
- Dai Z, Padture NP (2023) Challenges and opportunities for the mechanical reliability of metal halide perovskites and photovoltaics. *Nat Energy* 8:1319–1327
- Li N, Niu X, Chen Q, Zhou H (2020) Towards commercialization: the operational stability of perovskite solar cells. *Chem Soc Rev* 49:8235–8286
- National Laboratory of the Rockies (NLR). Best Research-Cell Efficiency Chart. <https://www.nlr.gov/pv/cell-efficiency>. Accessed 15 Apr 2026
- Ni Z et al (2020) Resolving spatial and energetic distributions of trap states in metal halide perovskite solar cells. *Science* 367:1352–1358
- Liu C et al (2023) Bimolecularly passivated interface enables efficient and stable inverted perovskite solar cells. *Science* 382:810–815
- Jiang Q et al (2019) Surface passivation of perovskite film for efficient solar cells. *Nat Photonics* 13:460–466
- Ni Z et al (2022) Evolution of defects during the degradation of metal halide perovskite solar cells under reverse bias and illumination. *Nat Energy* 7:65–73
- Cho KT et al (2017) Highly efficient perovskite solar cells with a compositionally engineered perovskite/hole transporting material interface. *Energy Environ Sci* 10:621–627
- Huang Y et al (2024) High-efficiency inverted perovskite solar cells via in situ passivation directed crystallization. *Adv Mater* 36:2408101
- Zhang Y et al (2025) Improving buried interface contact for inverted perovskite solar cells via dual modification strategy. *Adv Funct Mater* 35:2417575
- Tennyson EM, Doherty TAS, Stranks SD (2019) Heterogeneity at multiple length scales in halide perovskite semiconductors. *Nat Rev Mater* 4:573–587
- Shi P et al (2024) Micro-homogeneity of lateral energy landscapes governs the performance in perovskite solar cells. *Nat Commun* 15:9703
- Louis B et al (2025) In operando locally-resolved photophysics in perovskite solar cells by correlation clustering imaging. *Adv Mater* 37:2413126
- Howard JM, Lahoti R, Leite MS (2020) Imaging metal halide perovskites material and properties at the nanoscale. *Adv Energy Mater* 10:1903161
- Ma C et al (2023) Unveiling facet-dependent degradation and facet engineering for stable perovskite solar cells. *Science* 379:173–178
- Yao Y et al (2025) Oriented wide-bandgap perovskites for monolithic silicon-based tandems with over 1000 hours operational stability. *Nat Commun* 16:40
- Ma C et al (2022) Photovoltaically top-performing perovskite crystal facets. *Joule* 6:2626–2643
- Liu X et al (2024) Dominating (111) facets with ordered stacking in perovskite films. *Energy Environ Sci* 17:6058–6067

22. Sun X et al (2023) (111)-dominated perovskite films by antisolvent engineering. *Adv Mater* 35:2301115
23. Zhang X et al (2024) Preferred crystallographic orientation via solution bathing for high-performance inverted perovskite photovoltaics. *Adv Funct Mater* 34:2407732
24. Zhang P et al (2025) Carrier dynamics of facet-dependent heterogeneous interfaces in high-performance perovskite solar cells. *Adv Funct Mater*. <https://doi.org/10.1002/adfm.202422783>
25. Wang X, Han Z, Gao F, Luo C, Zhao Q (2022) Facet orientation and intermediate phase regulation via a green antisolvent for high-performance perovskite solar cells. *Sol RRL* 6:2100973
26. Gao F et al (2025) Perovskite facet heterojunction solar cells. *Joule* 9:101787
27. Xu J et al (2020) Triple-halide wide-band gap perovskites with suppressed phase segregation for efficient tandems. *Science* 367:1097–1104
28. Liu Z et al (2025) All-perovskite tandem solar cells achieving >29% efficiency with improved (100) orientation in wide-bandgap perovskites. *Nat Mater* 24:252–259
29. Zhou H et al (2024) Efficient and stable perovskite mini-module via high-quality homogeneous perovskite crystallization and improved interconnect. *Nat Commun* 15:6679
30. Li S et al (2024) Coherent growth of high-Miller-index facets enhances perovskite solar cells. *Nature* 635:874–881
31. Ni Z et al (2022) High grain boundary recombination velocity in polycrystalline metal halide perovskites. *Sci Adv* 8:eabq8345
32. Zhao L et al (2024) Surface-defect-passivation-enabled near-unity charge collection efficiency in bromide-based perovskite gamma-ray spectrum devices. *Nat Photonics* 18:250–257
33. Chen R et al (2022) Spatiotemporal imaging of charge transfer in photocatalyst particles. *Nature* 610:296–301
34. Meggiolaro D et al (2018) Iodine chemistry determines the defect tolerance of lead-halide perovskites. *Energy Environ Sci* 11:702–713
35. Ni Z et al (2024) Identification and suppression of point defects in bromide perovskite single crystals enabling gamma-ray spectroscopy. *Adv Mater* 36:2406193
36. Keeble DJ et al (2021) Identification of lead vacancy defects in lead halide perovskites. *Nat Commun* 12:5566
37. Wu X et al (2015) Trap states in lead iodide perovskites. *J Am Chem Soc* 137:2089–2096
38. Wang Y et al (2024) Homogenized contact in all-perovskite tandems using tailored 2D perovskite. *Nature* 635:867–873
39. Zhang F et al (2019) Enhanced charge transport in 2D perovskites via fluorination of organic cation. *J Am Chem Soc* 141:5972–5979
40. Zhang Y et al (2025) Reduction of nonradiative recombination at perovskite/C60 interface in inverted perovskite solar cells. *Adv Mater* 37:2500501
41. Wang X et al (2016) Characterization of conformation and locations of C-F bonds in graphene derivative by polarized ATR-FTIR. *Anal Chem* 88:3926–3934
42. Li M et al (2022) Oriented 2D perovskite wafers for anisotropic X-ray detection through a fast tableting strategy. *Adv Mater* 34:2108020

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